OVERVIEW OF ENERGY DATA ANALYTICS

ENERGY DATA ANALYTICS

Paolo Finardi

Energy & Finance Institute

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Agenda

1. Analytics in the AI Era
2. Data in Energy Companies
3. Energy Data Domains
4. Data Platform & Organization
5. The Analytics Toolkit

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Paolo Finardi

Head of
Data Science & AI

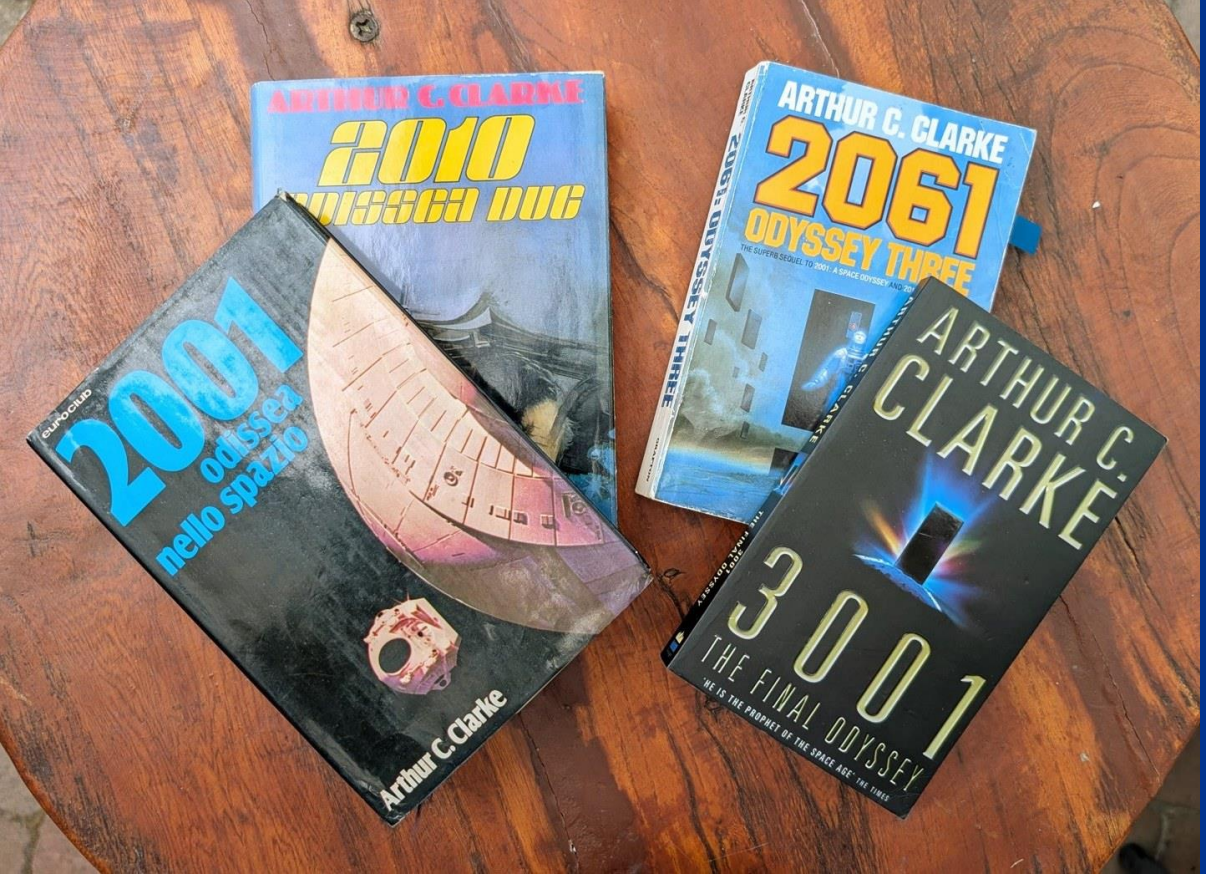
- **Leads AI Strategy** and a team of 30+ Data Scientists across the Retail, E-Mobility, and Renewables sectors.
- **10+ years of experience** focused on scalable ML solutions in Forecasting and Energy Optimization.
- **Active community contributor**: Member of the Databricks Product Advisory Board and Organizer of PyData Milan.

www.linkedin.com/in/paolo-finardi-38b25096

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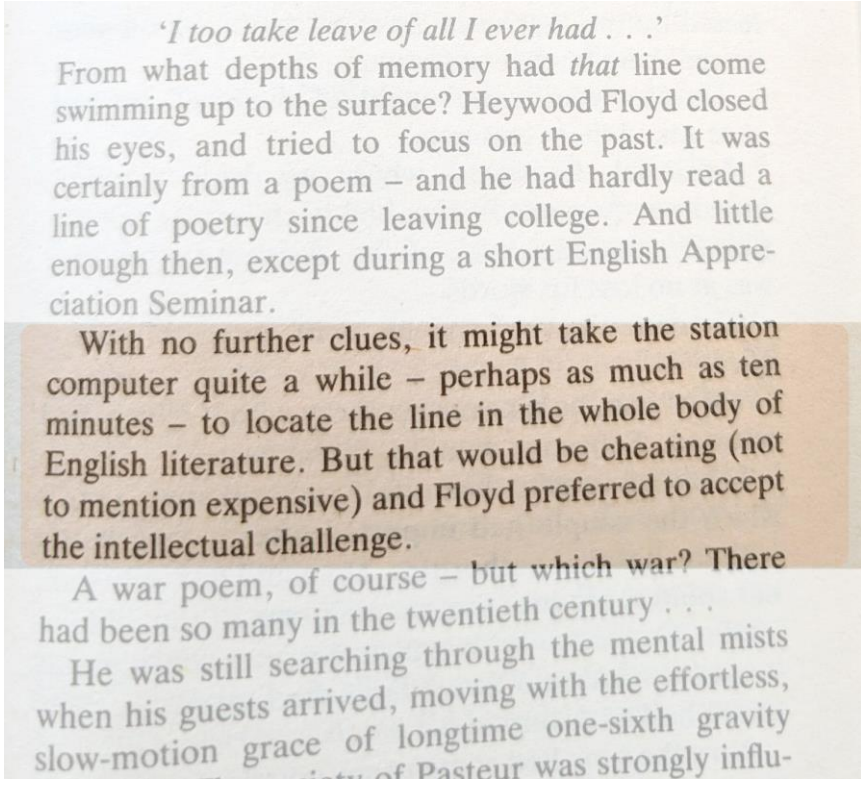
01 Analytics in the AI Era

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SPACE ODYSSEY · 1968–1997

How did we imagine the future?



'I too take leave of all I ever had . . .'

From what depths of memory had *that* line come swimming up to the surface? Heywood Floyd closed his eyes, and tried to focus on the past. It was certainly from a poem – and he had hardly read a line of poetry since leaving college. And little enough then, except during a short English Appreciation Seminar.

With no further clues, it might take the station computer quite a while – perhaps as much as ten minutes – to locate the line in the whole body of English literature. But that would be cheating (not to mention expensive) and Floyd preferred to accept the intellectual challenge.

A war poem, of course – but which war? There had been so many in the twentieth century . . .

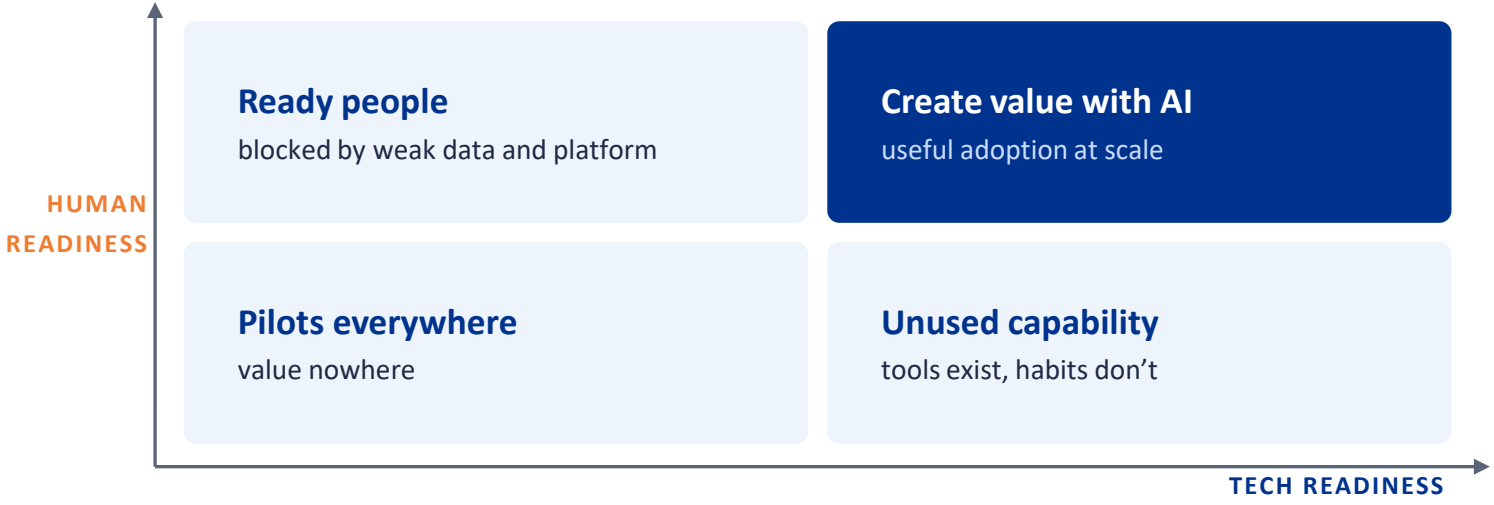
He was still searching through the mental mists when his guests arrived, moving with the effortless, slow-motion grace of longtime one-sixth gravity

ARTHUR C. CLARKE
2061: ODYSSEY THREE · 1987

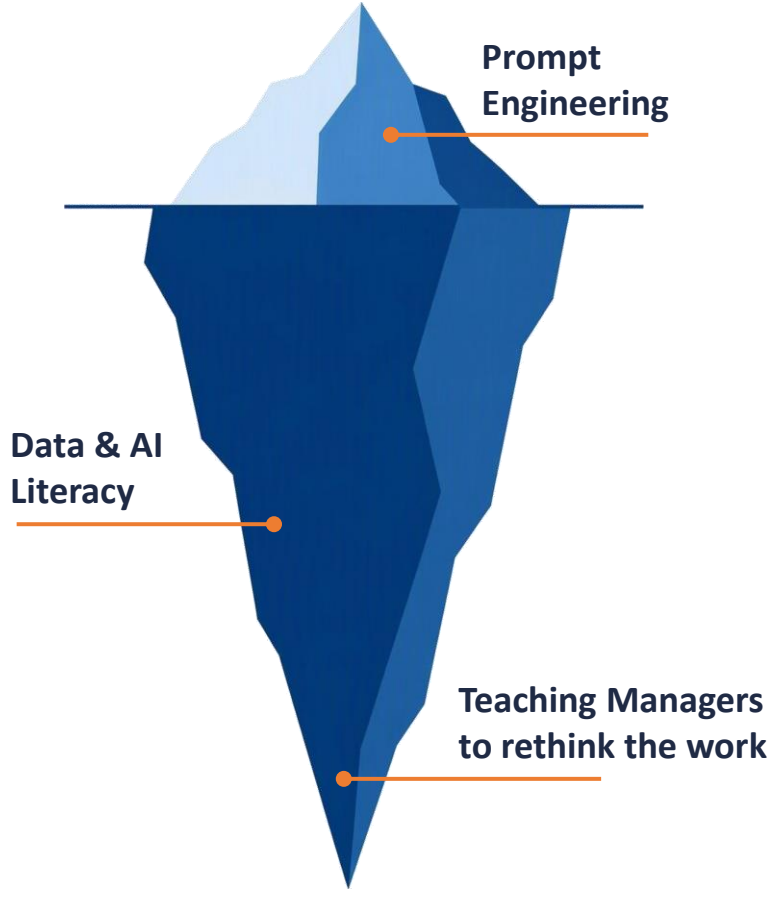
"perhaps as much as ten minutes to locate a line in the whole body of English literature"

AI readiness: two axes, one outcome

AI creates value when capability and adoption move together.



How do we improve AI human readiness?



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Three ingredients, one direction

1 **Process over use case**

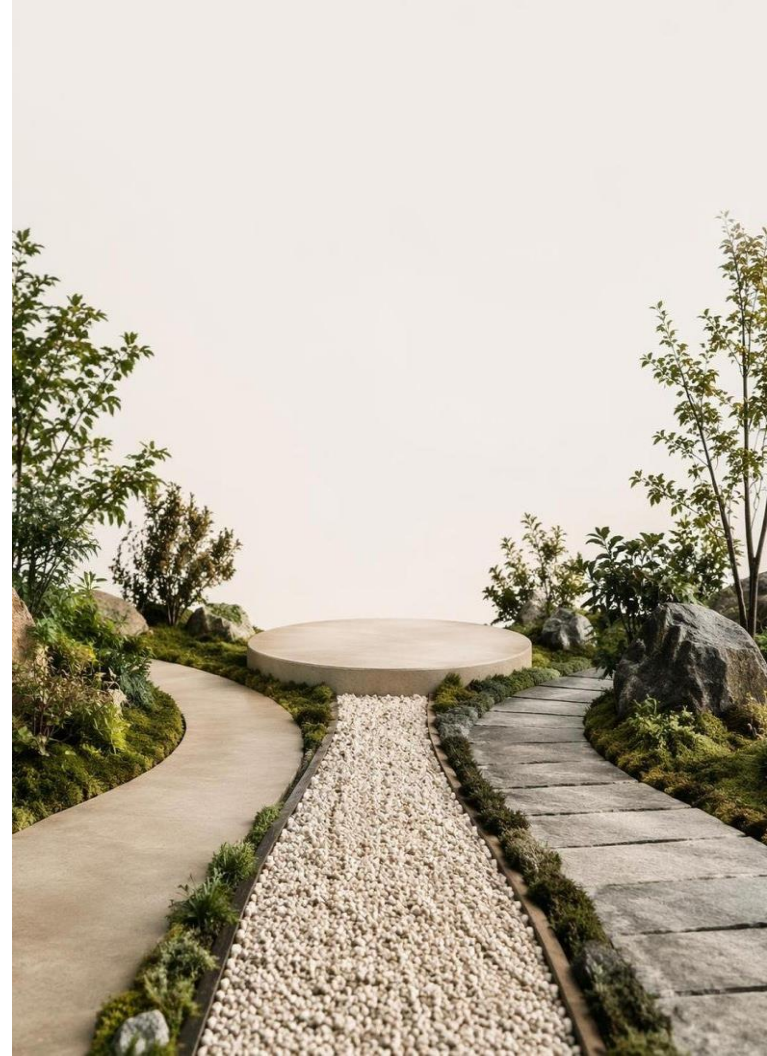
Redesign the work instead of patching it

2 **Employee empowerment**

Put the tools in the hands of who does the work

3 **Job impact**

Plan for superworkers, and reward them



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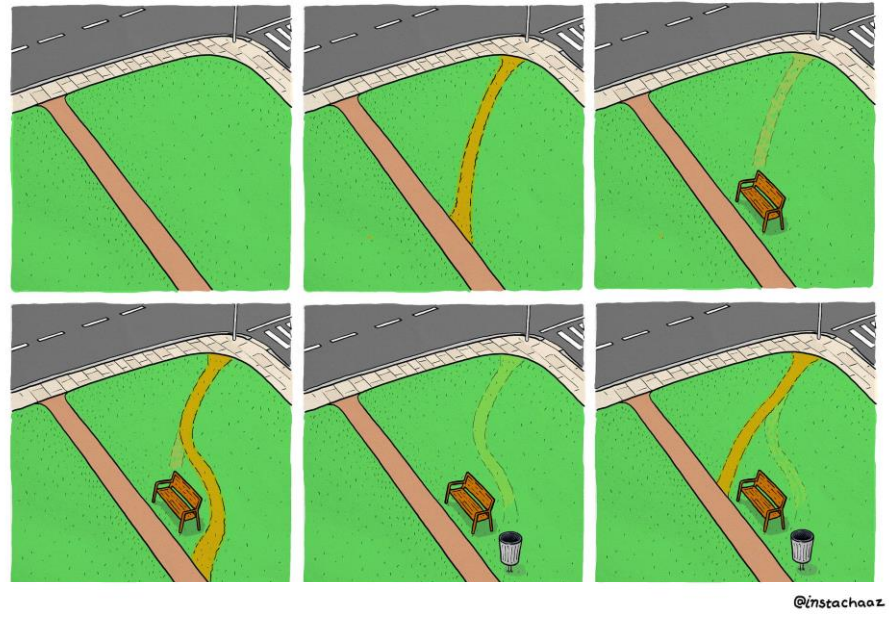
Don't automate the old process. Redesign it.

Ask what you'd build from scratch — not how to speed up what's there.

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They'll show you the shortcut.

Give them the tools — then build for the path they actually take.



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We tend to overestimate the effect of a technology in the short run and underestimate the effect in the long run.

— Roy Amara

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BBC

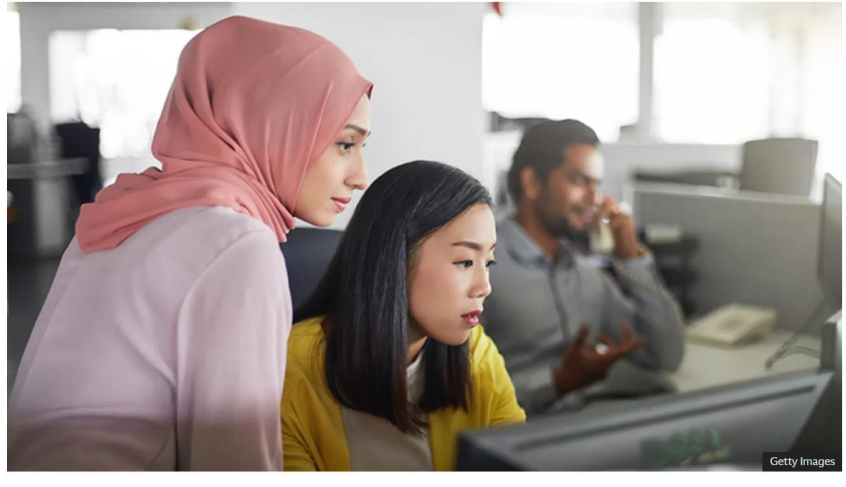
Home News Football 2026 Business Technology Health Culture Arts Travel Earth Sport Audio Video Live

AI could replace equivalent of 300 million jobs - report

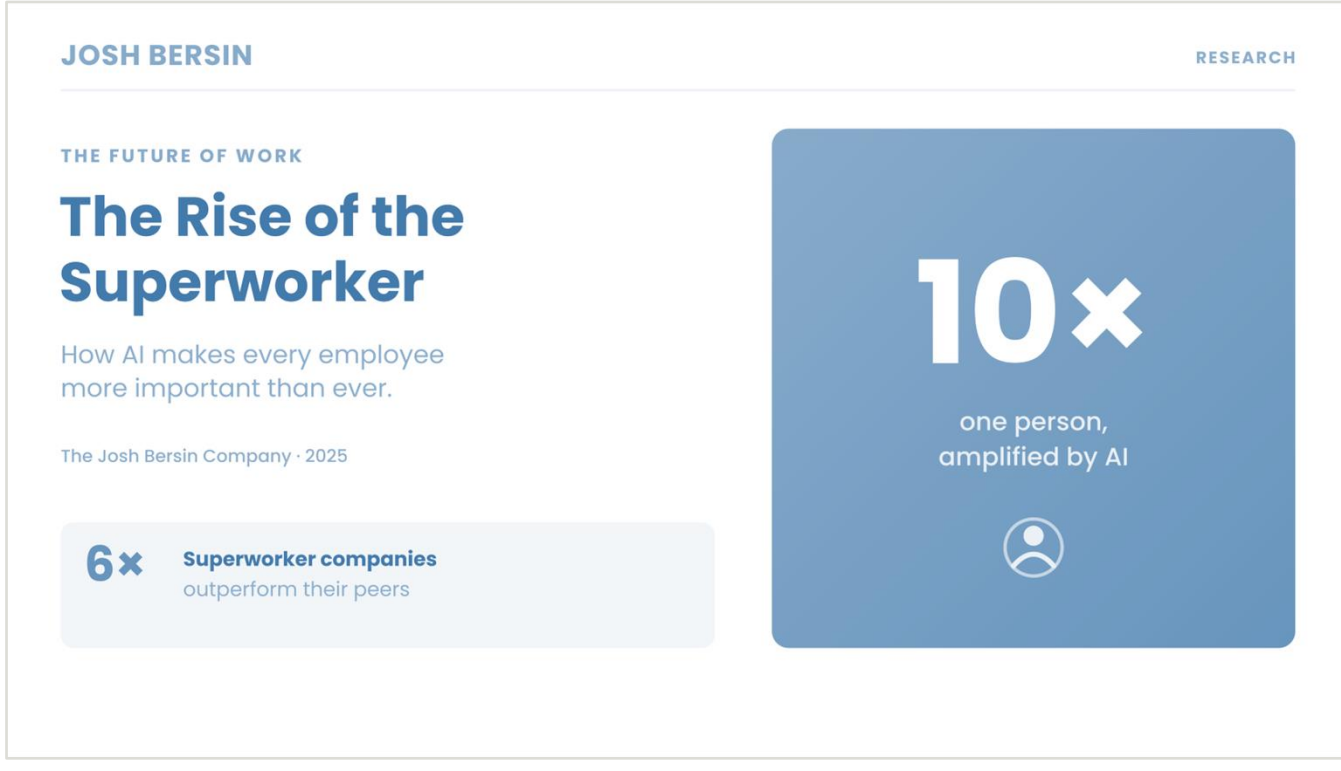
28 March 2023

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Chris Vallance
Technology reporter



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JOSH BERSIN
RESEARCH
THE FUTURE OF WORK
The Rise of the Superworker
How AI makes every employee more important than ever.
The Josh Bersin Company · 2025
6×
Superworker companies
outperform their peers
10×
one person, amplified by AI

Energy Data Analytics

1 Data in Energy – Introduction

Data, platforms, tools

YOU ARE HERE

2 Solar Power Generation

PV data science lifecycle and forecasting

3 Wind Energy Forecasting

Physics and data-driven models, PyWake

4 Analytics for BESS

Battery data, KPIs, simple models

5 Trading Analytics

Market fundamentals and a trading strategy

6 Wholesale BESS Optimization

Dispatch with MILP and Pyomo

7 Energy and Agentic AI

From ML to agents: a market briefing agent

8 Data Visualization

SCADA in Power BI: power curves, underperformance

9 Predictive Maintenance

Turbine anomalies and maintenance economics

10 Machine Learning

Customer clustering and demand response

11 Energy Loss Forecasting

Technical losses with SARIMAX

12 Capstone Simulation

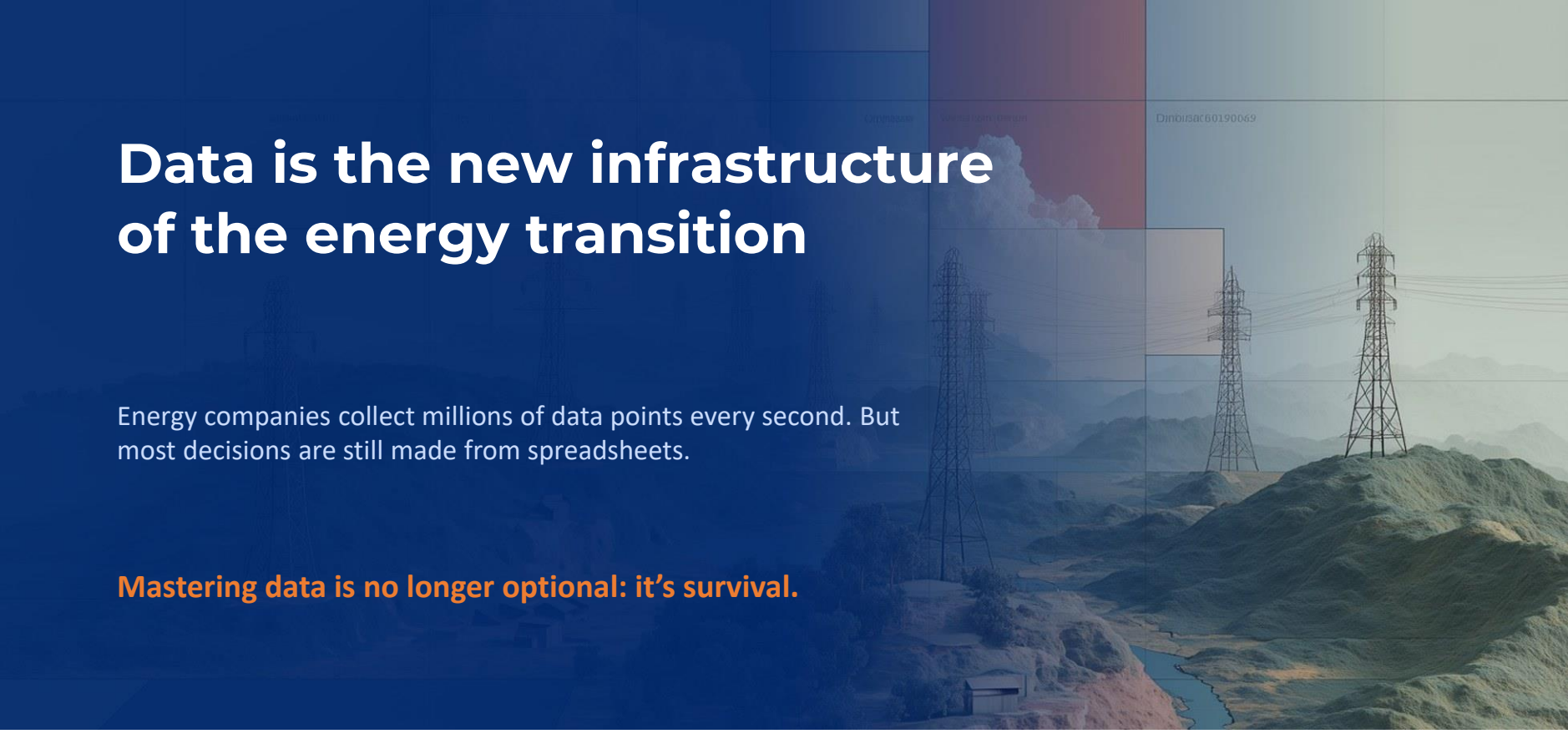
Forecasting, markets and finance, end to end

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02 | Data in Energy Companies

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Data is the new infrastructure of the energy transition



Energy companies collect millions of data points every second. But most decisions are still made from spreadsheets.

Mastering data is no longer optional: it's survival.

Why the stakes are high

86%

of global energy supply still comes from fossil fuels (2025)

Energy Institute, Statistical Review 2026

32% → 43%

renewables' share of global electricity, 2024 → 2030

IEA, Renewables 2025

573 h

of negative day-ahead prices in Germany in 2025, +25% vs 2024

Bloomberg, January 2026

So what: more variable supply means more volatility. Forecasting, flexibility and data are where the margin is made.

Three ways data creates value

Optimization

Do the same with less: cost, risk, errors

- Better forecasts, lower imbalance costs
- Predictive maintenance, less downtime

New opportunity

New revenue streams built on data

- Flexibility and BESS trading
- Dynamic tariffs and energy services

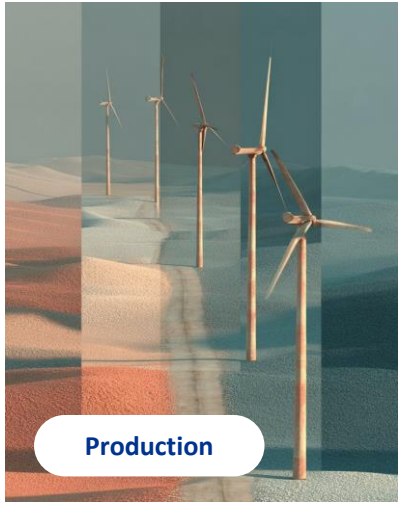
Strategic asset

Data maturity is priced in

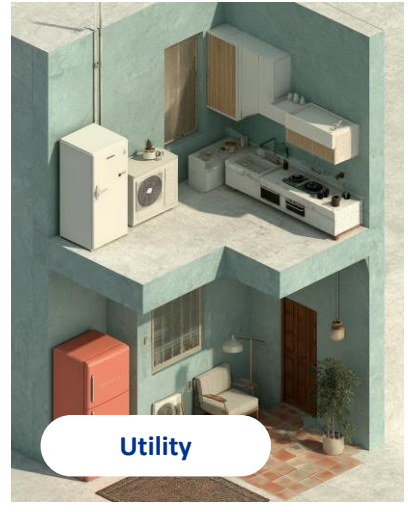
- Investors reward data platforms that mitigate operational risk when setting the multiple
- Better operational data reduces production uncertainty: better financing terms

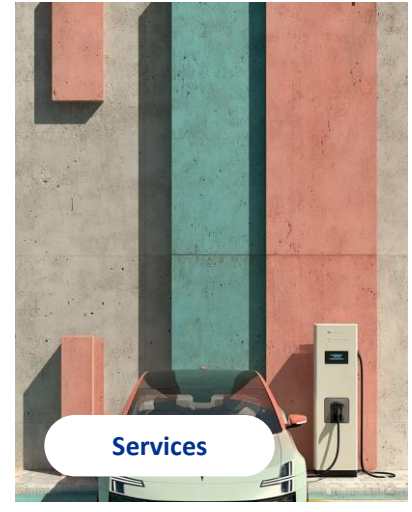
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Inside an energy company









Generation

Market

Retail

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03 | Energy Data Domains

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Three data domains, one value chain



Renewable Generation

How much will we produce, and are our assets performing?

- SCADA & sensors
- Weather, measured & forecast
- Asset master data
- Maintenance & alarms



Energy Market

What do we buy and sell, at what price, and how do we hedge our customers' demand?

- Prices & forward curves
- Market fundamentals
- Portfolio position & hedges
- Counterparty & credit



Retail

Who are our customers, what do they consume, and do they pay?

- Customer & contract
- Metering & consumption
- Billing & payments
- Interactions & campaigns

Different clocks: SCADA every few minutes, markets every 15 minutes, bills every month. Aligning them is the first analytics job.

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Renewable generation data

Data	What it contains	Granularity	Typical pitfalls
SCADA & sensors	Active power, availability, inverter / turbine status, temperatures	1–15 min	Gaps, frozen values, sensor drift, unflagged curtailment
Weather	Irradiance, wind speed & direction, temperature: measured on site and forecast (NWP)	15 min – 1 h	Mixing forecast and actuals, forecast issue time, site vs grid point
Asset master data	Capacity, location, tilt / azimuth, turbine model, power curve	Static	Outdated after repowering, inconsistent IDs across systems
Maintenance & alarms	Work orders, alarm codes, component replacements	Event	Free text, missing timestamps, alarm floods
Storage (BESS)	State of charge, charge / discharge, cycles, efficiency	1–15 min	Sign conventions, SoC calibration drift

In this course

Solar PV forecasting | Wind forecasting | SCADA visualization | Predictive maintenance | BESS

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Energy market data

Data	What it contains	Granularity	Typical pitfalls
Prices	Day-ahead, intraday and imbalance prices by bidding zone	15 min – 1 h	Time zones & DST, the 2025 switch of EU day-ahead to 15 min
Forward curves & products	Futures (base / peak, month to year), PPAs	Daily	Product and delivery-period mapping, illiquid tenors
Market fundamentals	Load, renewable output, fuel & CO_2 prices, plant outages	15 min – daily	Revisions after publication, forecast vs actual
Portfolio position & hedges	Customer demand forecast, own generation, signed contracts	15 min – monthly	Different systems and granularities, late customer data
Counterparty & credit	Exposures, ratings, collateral, limits	Daily	Stale ratings, limits scattered across systems

Typical sources: power exchanges, ENTSO-E Transparency Platform, brokers, ETRM systems

In this course Power market & trading | BESS optimization | AI market briefing agent | Capstone

Retail data

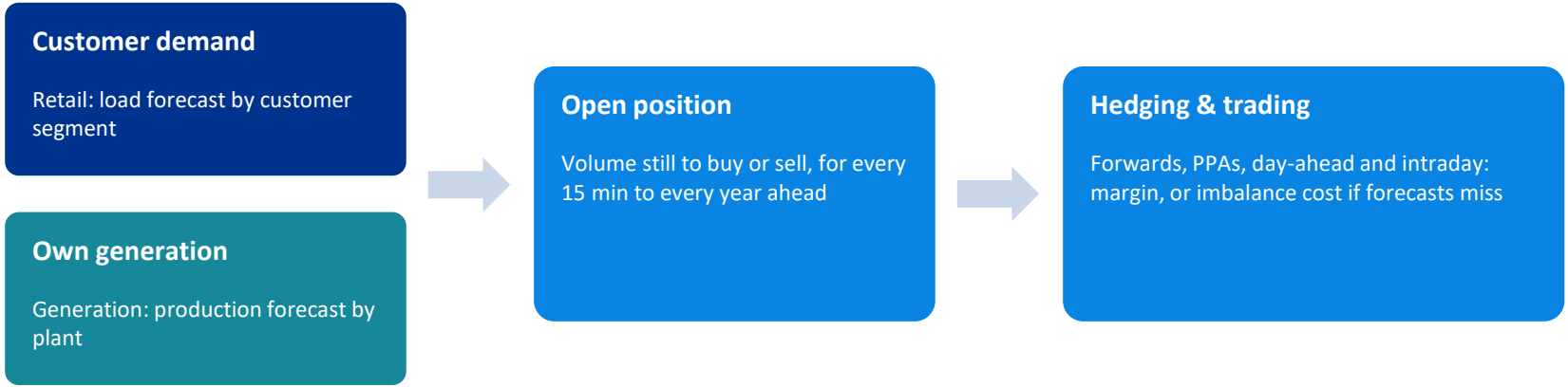
Data	What it contains	Granularity	Typical pitfalls
Customer & contract	Master data, offer and tariff, contract dates, metering point IDs	On change	Duplicates, personal data (GDPR), offer codes changing over time
Metering & consumption	Smart-meter readings (power), meter reads (gas), load profiles	15 min – monthly	Estimated vs actual reads, gaps, switching in and out
Billing & payments	Invoices, payment dates, arrears, payment method, credit checks	Monthly / event	Rebilling and adjustments, partial payments
Interactions & campaigns	Contact center, app, complaints, campaigns and responses	Event	Free text, channels not linked, marketing consent
Competitor offers	Competitor tariffs, price-comparison data	Weekly – monthly	Offers hard to compare like for like

The customer demand forecast built here drives the market position: see next slide

In this course Customer segmentation & demand response | Capstone

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Where the domains meet



Open position(t) = Customer demand(t) − Own generation(t) − Contracted volumes(t)

Takeaway: a forecast error in one domain shows up as a cost in another. This is why data silos are expensive.

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Breaking the walls: data unification

1

Overcoming Data Silos

Isolated data limits visibility and hampers quick decision-making, creating significant barriers to organizational agility.

2

Integrated Analytics for Insights

Combining different data sources enables deeper analytics, leading to smarter and more holistic business strategies.

3

Governance and Standards

Establishing clear standards and governance ensures consistent, trusted data across the organization.

4

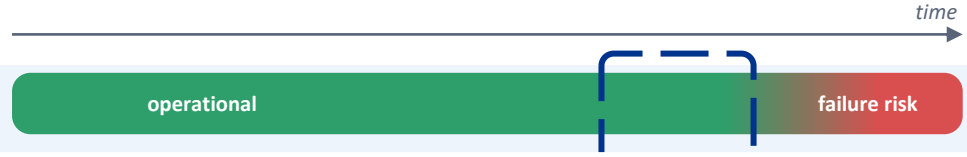
Ensuring Interoperability

Technology solutions should enable seamless data flow across different platforms and teams for maximal value.

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Wind case: the maintenance window

What we need to know	Data from
Turbine health	SCADA + physics model + ML



04 | Data Platform & Organization

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Evolution of the data platform

maturity

Legacy systems

What it is

On-prem databases, Excel reports

Limit

Data siloed by department, no real-time access

Data warehouse

What it is

Structured, SQL-based, finance and reporting focus

Limit

Slow to adapt, expensive, limited unstructured data

Data lake

What it is

Stores everything, structured and raw, cheaply

Limit

Chaos without governance: “data swamp” risk

Lakehouse

What it is

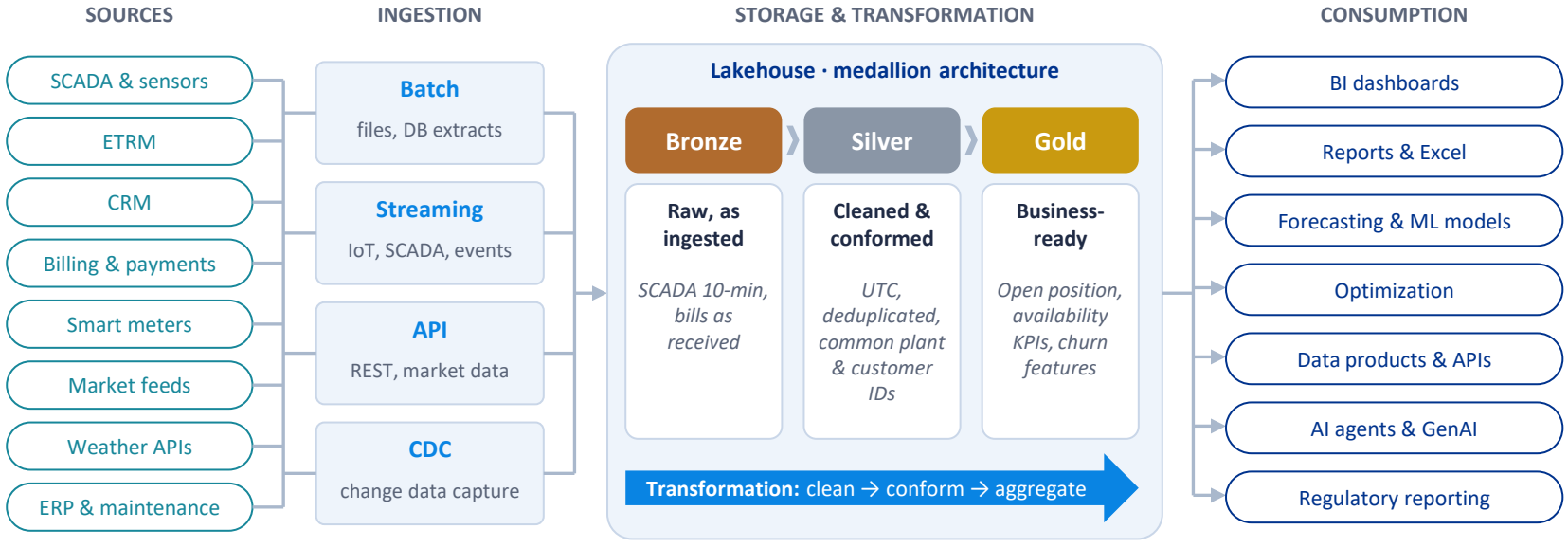
Unified governance, scalable compute, structured + unstructured

Limit

Needs a culture shift and shared ownership to deliver

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Data platform building blocks



Governance across all layers: catalog · data quality · security & access · lineage

Data teams: the translators

Toward the business

- Idea generation
- Spread a data-driven culture
- Understand processes to propose improvements
- Make the business able to propose its own AI use cases

Data team

- Data analyst
- Data scientist
- Data engineer
- Data governance specialist

Toward IT

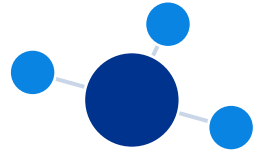
- Integrate data tools into existing systems
- Evolve systems for analytics and AI
- Evolve cyber security and data protection

Scaling the organization



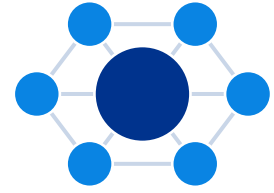
Data hub

Central team does everything



Hub + business spokes

Central team plus data people in the business



Federated / data mesh

Domains own their data, shared governance

maturity

What makes it scale: change management

Data culture

From reporting to data-driven decisions

Sponsorship

Engage management with vision and results

Governance

Define ownership, tools and frameworks

Learning

Cross-skill transfer business ↔ data

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“The Data Platform is not an IT project — it’s the foundation for the intelligent, decarbonized energy company.

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05 | The Analytics Toolkit

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Method first: the lifecycle



Business understanding

What problem are we solving? What metric defines success?

Output: Business canvas

Data understanding

What data do we have? Is it reliable?

Output: Data inventory, first exploration

Data preparation

How do we clean and structure it?

Output: Ready-to-model dataset

Modeling

Which algorithm or approach?

Output: Models vs a baseline

Evaluation

Does it work? Is it stable and usable?

Output: Metrics, backtest results

Deployment

How do we deliver the value?

Output: Dashboard, decision support, automation

Iterative: we move back and forth between phases.

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Four tools, different jobs



Excel

Best for

Quick checks, small datasets, financial models everyone can read

Watch out

Max 1,048,576 rows: one year of 1-min SCADA for two turbines already exceeds it



Python

Best for

Large time series, forecasting, optimization, automation, reproducible analysis

Watch out

Needs coding skills and versioned code



BI (e.g. Power BI)

Best for

Monitoring, KPIs, sharing results with the business

Watch out

Weak for modeling; business logic hidden inside dashboards



AI assistants

Best for

Writing and explaining code, first-draft analysis, documentation

Watch out

Plausible but wrong answers: you need the basics to check them

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Python stack and open data

Need	Library	Where in the course
Data handling	pandas, NumPy	All hands-on sessions
Time series	statsmodels, Prophet	Solar PV forecasting, technical losses
Machine learning	scikit-learn	Customer segmentation, anomaly detection
Wind modeling	PyWake	Wind forecasting
Optimization	Pyomo	BESS optimization
Dashboards	Plotly, matplotlib	SCADA visualization



Open data to practice on

ENTSO-E Transparency Platform
EU load, generation, prices

Copernicus ERA5 / Open-Meteo
Weather, historical and forecast

New European Wind Atlas
Wind resource and site assessment

Energy-Charts (Fraunhofer ISE)
German power market and generation

Kaggle Solar Power Generation
Dataset of the solar hands-on

Q&A

Thank you

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